

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.



Dataset at a Glance

3,900

Total Purchases

Transactional records across all categories

18

Columns

Demographics, purchase details, and behavior

37

Missing Values

In Review Rating column only

25

Product Types

Unique items purchased

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount, Promo Code, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

Loading & Exploration

Imported dataset via `pandas`; used `df.info()` and `.describe()` for structure and summary statistics.

Cleaning

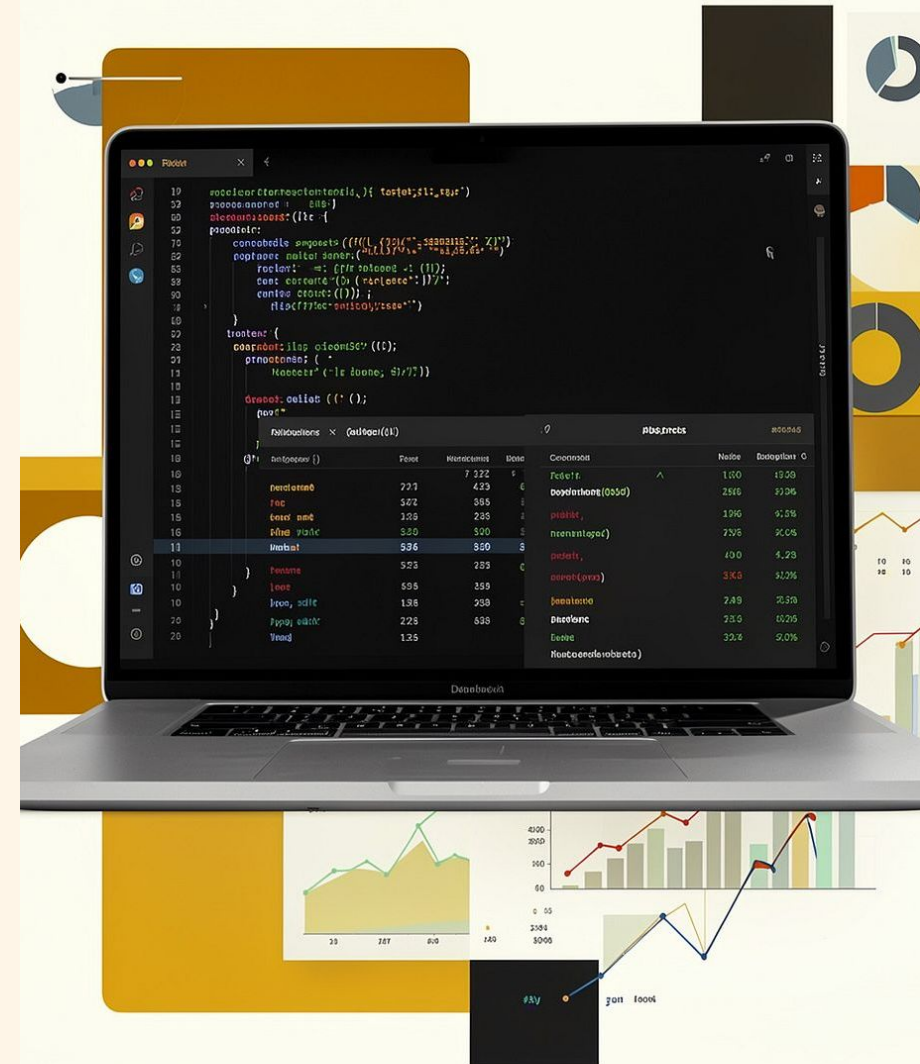
Imputed missing Review Ratings using the **median per product category**; renamed columns to `snake_case`.

Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used` column.

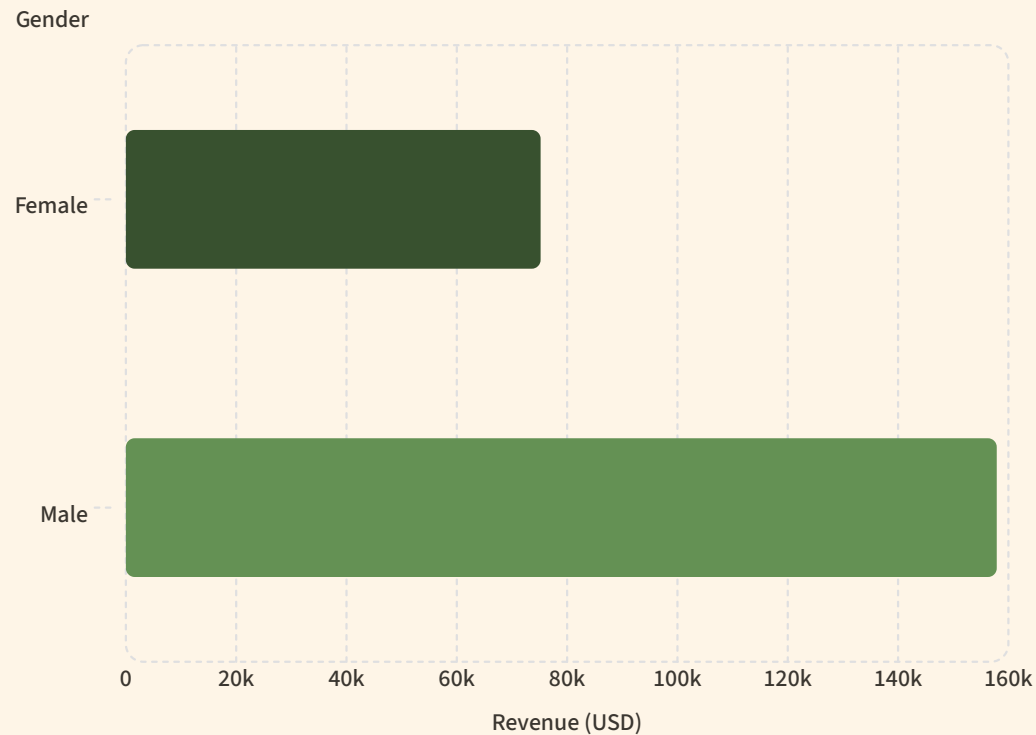
Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for SQL-based analysis.

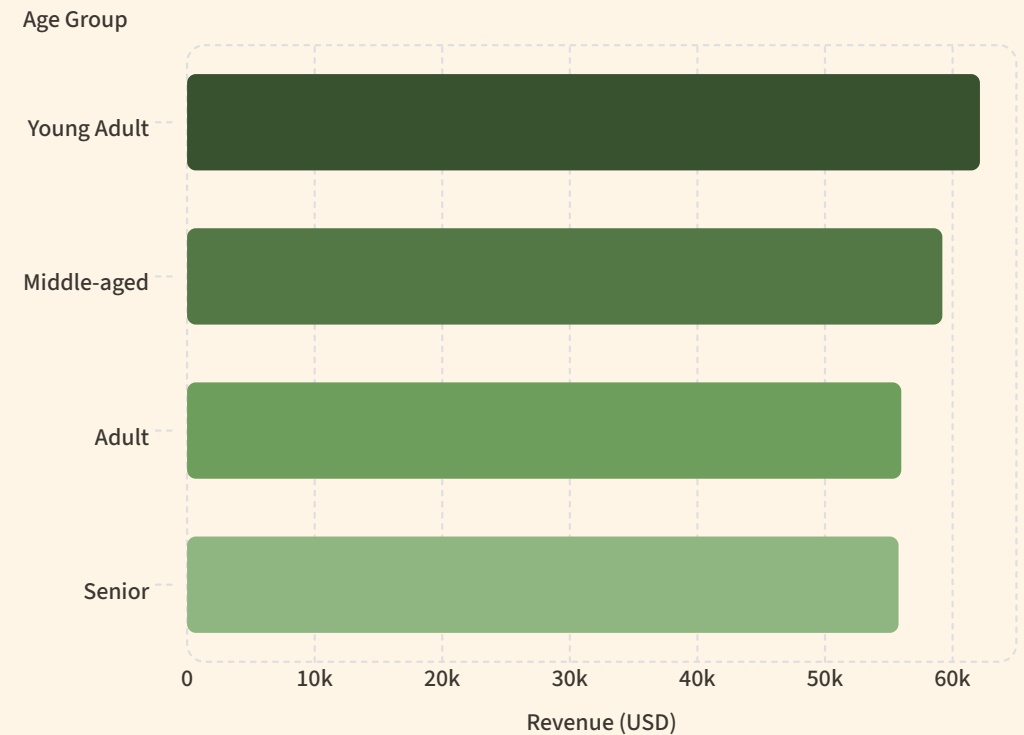


Revenue Insights: Gender & Age

Revenue by Gender

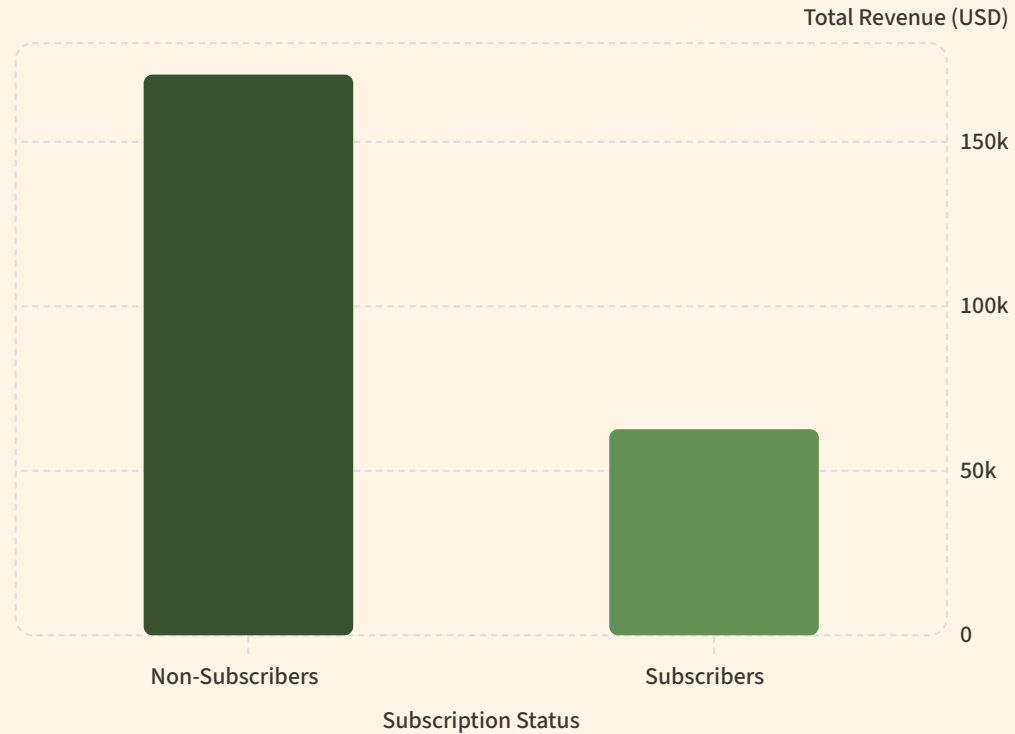


Revenue by Age Group



Male customers generated over **2x the revenue** of female customers. Young Adults led age groups, though revenue was relatively evenly distributed across all segments.

Subscribers vs. Non-Subscribers



Key Findings

Non-subscribers drive **73% of total revenue**, despite nearly identical average spend per transaction.

Non-Subscribers

2,847 customers · Avg spend: **\$59.87** · Revenue: \$170,436

Subscribers

1,053 customers · Avg spend: **\$59.49** · Revenue: \$62,645



Customer Segmentation



Loyal

3,116 customers — the dominant segment with high repeat purchase history.



Returning

701 customers — engaged but not yet fully loyal; key conversion target.



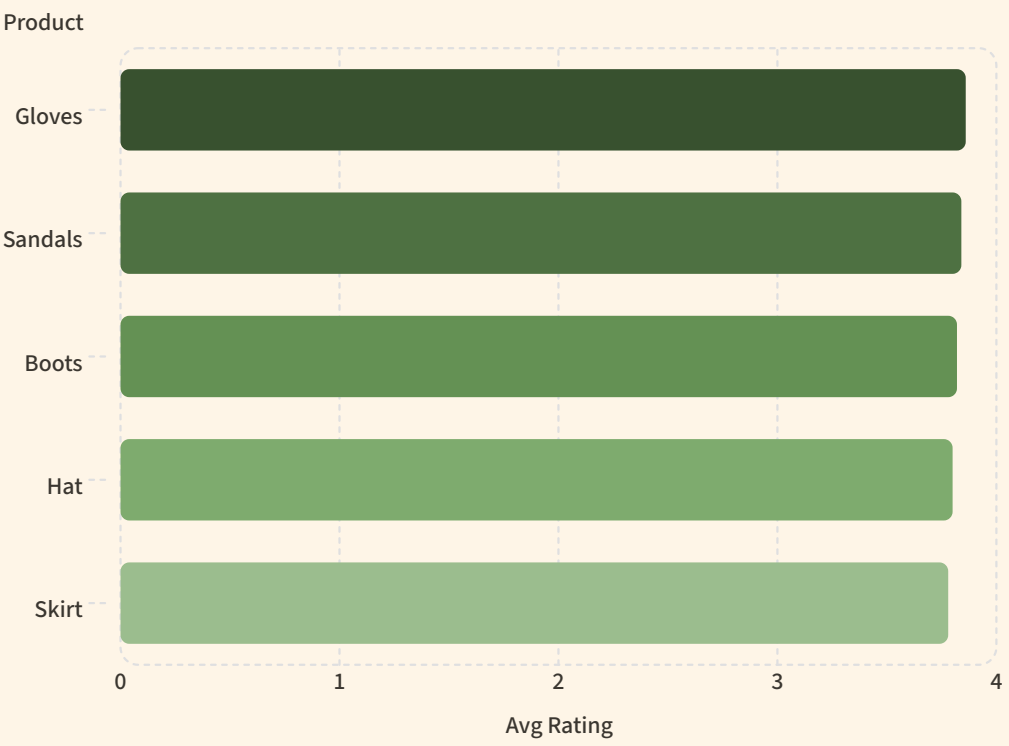
New

83 customers — smallest segment; opportunity for onboarding and retention.

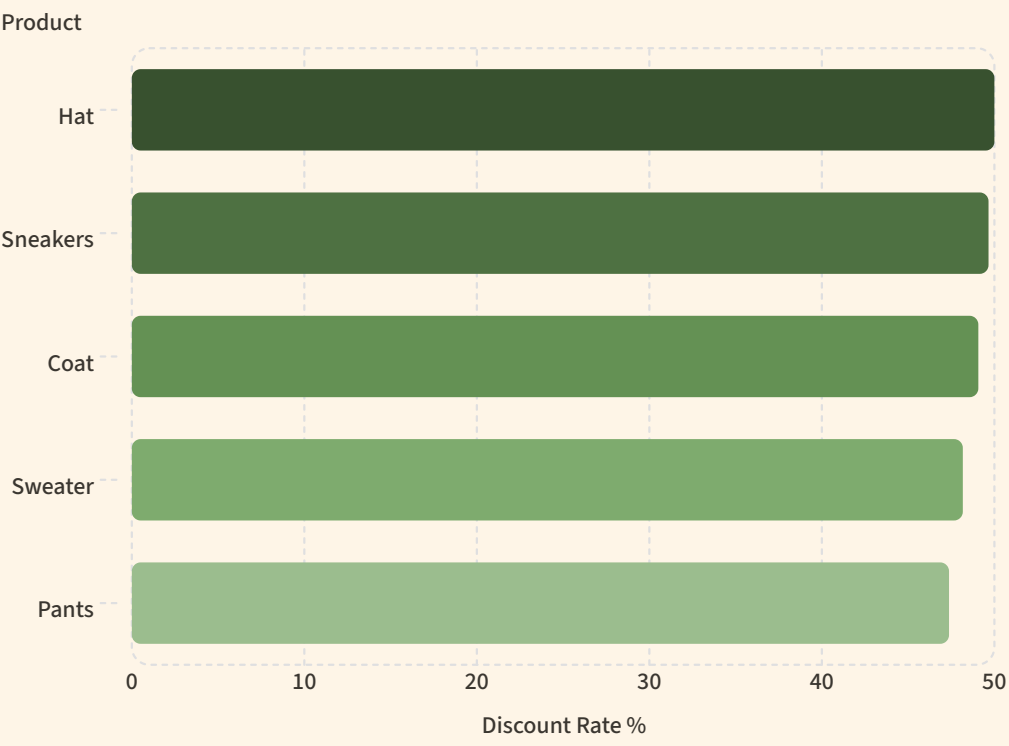
Of customers with **>5 purchases**, 958 are subscribers vs. 2,518 non-subscribers — suggesting room to convert repeat buyers into subscribers.

Top Products & Discount Behavior

Top 5 Products by Review Rating



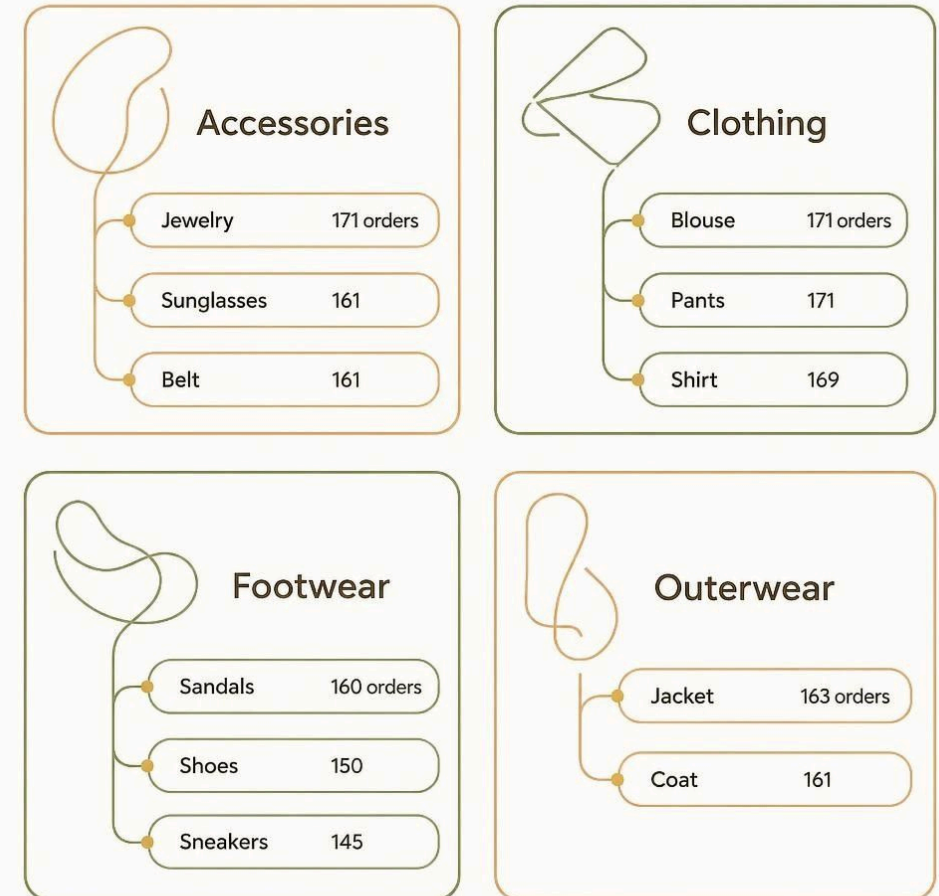
Most Discount-Dependent Products



839 customers used discounts and still spent above the average purchase amount — indicating strong price sensitivity among higher-value buyers.

Top Products by Category

The most purchased items within each category reveal clear consumer preferences. **Clothing** dominates overall with 1,737 purchases, led by Blouse (171 orders). In **Footwear**, Sandals top the list (160), followed by Shoes (150) and Sneakers (145). **Accessories** are led by Jewelry (171), while **Outerwear** sees Jacket (163) and Coat (161) as top sellers.



Power BI Dashboard

An interactive **Customer Behavior Dashboard** was built to visualize all insights with filters for Subscription, Gender, Category, and Shipping Type.



3.9K Customers
Total customer count



\$59.76 Avg
Average purchase amount



3.75 Avg
Average review rating



Key Charts
Revenue by category, age group, and subscription status



Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits to convert the large non-subscriber base.

→ Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

→ Review Discount Policy

Balance sales boosts with margin control — 839 discount users still spent above average.

→ Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.

→ Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.